

Shankar Kumar Suddamalla

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Summary

Backend Java Developer with strong experience in developing and supporting enterprise-scale applications using Java, Spring Boot, and Microservices architecture. Hands-on experience in designing Restful APIs, Implementing Kafka-based asynchronous communication, Processing XML/JSON messages, and developing robust backend services for high-volume transaction processing. Skilled in Spring Data JPA, SQL, Git, Maven and production support, with a solid understanding of distributed systems, event-driven architecture, and software design principles.

Work Experience

Full-Stack Engineer – Tata Consultancy Services, India

May 2025 – Present

- Developed and enhanced enterprise-grade backend services using Java, Spring Boot, and Microservices to support secure, high-volume payment processing workflows.
- Implemented RESTful APIs, business validations, XML message processing, and data transformations to ensure accurate and reliable transaction execution across distributed systems.
- Integrated Apache Kafka for asynchronous event-driven communication, enabling efficient message exchange and improving system scalability and reliability.
- Utilized Apache Ignite distributed caching to improve application performance, reduce database access, and support low-latency transaction processing in enterprise applications.
- Resolved production Issues through debugging, log Analysis, root cause analysis and performance optimization.
- Technologies: Java, Spring Boot, Spring Framework, Microservices, REST APIs, Apache Kafka, Apache Ignite, PostgreSQL, Gradle, Postman, Spring Cloud, Spring Data JPA

Technical Skills

Languages: Java, JavaScript, Python

Frameworks: Spring, Spring Boot, Spring Data JPA, Spring Security, Spring Cloud

Messaging and Caching: Apache Kafka, Apache Ignite **Architecture:** Microservices, RESTful APIs, Event-Driven Architecture **Databases:** PostgreSQL, MySQL

Tools: Git, GitHub, AWS, Kafka, JMeter, Gradle, Maven, Postman

Education

Sastra Deemed University

2020–2024

Computer Science Engineering (GPA 7.77)

BTech

Narayana Junior College

2018–2020

Intermediate Education MPC (marks:977)

Balaji High School

2016–2018

Secondary Education (CGPA:10)

Personal Projects

Inventory Management System

- Designed and developed a full-stack Inventory Management System using Java, Spring Boot, Angular, and MySQL to automate inventory tracking, product management, and stock operations.
- Built secure RESTful APIs for managing products, categories, suppliers, purchases, and sales, implementing input validation, exception handling, and business rules.

- Implemented role-based access control (RBAC) to provide secure access for administrators and users, ensuring authorized access to inventory operations.
- Developed inventory workflows for stock addition, stock deduction, and real-time inventory updates, improving data accuracy and operational efficiency.
- Integrated the Angular frontend with Spring Boot backend, documented APIs using Swagger/OpenAPI.

Enterprise-Grade Distributed Load Balancer

- Designed and developed an enterprise-grade Layer 7 Intelligent Load Balancer using Java and Spring Boot to distribute client requests efficiently across multiple backend services.
- Implemented multiple load-balancing algorithms including Round Robin, Least Connections, Weighted Round Robin, and IP Hash with configurable routing strategies.
- Built active health monitoring and automatic failover mechanisms to detect unhealthy service instances and route traffic only to healthy nodes, improving system availability.
- Developed a scalable request routing engine with support for service discovery, dynamic backend registration, rate limiting, and request filtering to simulate production-grade traffic management.
- Integrated Docker, Kubernetes, Prometheus, and Grafana to deploy, monitor, and visualize load balancer performance, latency, throughput, and backend service health.

Enterprise E-Commerce Platform using Microservices

- Designed and developed an enterprise-grade e-commerce platform using Java, Spring Boot, and Microservices Architecture, enabling scalable and modular business services.
- Built RESTful microservices for product catalog, inventory, shopping cart, order management, payment processing, and user management, following domain-driven service boundaries.
- Implemented Apache Kafka for asynchronous communication between microservices, ensuring reliable event-driven order processing, inventory updates, and notification workflows.
- Integrated Redis for distributed caching and optimized database interactions to improve application performance and reduce API response times under high traffic.
- Containerized services using Docker and orchestrated deployments with Kubernetes, while implementing API Gateway, centralized configuration, service discovery, and monitoring for a cloud-native architecture.